

DBMS LAB 7

Instructions:

- Submit one **.pdf**, one **.sql**, and one **.ipynb** file.
 - The pdf file should only contain the query and outputs
 - All the python codes should be in one .ipynb (Jupyter Notebook) file.
 - The name should be **<roll no>_Assignment7.ipynb/.pdf/.sql**
 - Write the questions before the answers.
 - Complete the exercise before 4.45 PM and submit the files in moodle.
 - Marks Distribution for both Postgres and python (query/code + output)
 - **Total Marks = 20**
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Note: Use below link to restore the dvdrental database before answering the questions.

Download tar file from: <https://github.com/imkumaraju/dvdrenat-sample-database>

- 1) Write a procedure named **process_rental_payment** that takes p_customer_id, p_rental_id, and p_amount as input.
 - Verify that the provided customer_id exists. (1)
 - Verify that the rental_id exists, belongs to the provided customer, and does not already have an associated payment (is unpaid). (2)
 - If valid, insert a record into the payment table using a default staff_id of 1, and the current timestamp for the payment_date. (1)
 - Ensure proper error handling and transaction control. (1)
- 2) Write a transaction in the that performs the following operations:
 1. Start a transaction and update the **rental_rate** of the film with film_id = 10 to 3.99. (1)
 2. Create a **SAVEPOINT** named **sp_film_update**. Then update the **replacement_cost** of the same film to 25.99. (1)
 3. Assume the second update was incorrect. **Rollback to the savepoint** so that only the first update remains, and then **commit the transaction**. (1)
- 3) Write a procedure named **print_numbers** that uses a **WHILE loop** to display numbers from **1 to 10** using RAISE NOTICE. (3)

NOTE: Complete the rest of the assignment using python

- 4) Establish a connection to the database. (1)
- 5) Find the title of films that have never been rented (3)
- 6) Find the **first name, last name, and email** of customers who live in the **city 'London'**. (2)
- 7) Given a **staff_id**, find the **total number of rentals processed by that staff member**. (3)